

**VIT**Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

240101069

Continuous Assessment Test (CAT) – II SEPTEMBER 2026

Programme	: B.Tech.	Semester	: Fall Semester 2026-27
Course Code & Course Title	: BECE302L & Control Systems	Class Number	: CH2026270102555 CH2026270102550 CH2026270102552 CH2026270102548
Faculty	: Dr Raju Patel Dr. Dheeren Ku Mahapatra Dr Manmohan Sharma Dr Akash Kumar Pradhan	Slot	: B2+TB2
Duration	: 90 Minutes	Max. Mark	: 50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary for Question 3 (Root Locus).
- Use Semi log graph sheet supplied from the exam cell as necessary for Question 4 (Bode Plot).
- Use Polar graph sheets supplied from the exam cell as necessary for Question 5 (Polar plot).
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1.		For the unity-feedback system with OLTF $G(s)$, determine the value of K. $G(s) = \frac{K(s+3)}{s^2(s+7)}$ If the input is $r(t) = 10t^2u(t)$, and the desired steady-state error is 0.061 for this input. Determine the corresponding value of Acceleration Error Constant (K_a) for the obtained value of K.	10	2	2
2.		Use the Routh-Hurwitz criterion to determine the location of roots on the s-plane and hence the stability for the system represented by the characteristic equation, $s^6 + s^5 + 3s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$	10	3	3
3.		Sketch the root locus plot for the system having open-loop transfer function is given by $G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+13)}$	10	3	3
4.		The transfer function of a control system is represented by, $G(s)H(s) = \frac{1500(s+20)}{s^2(s+5)(s+300)}$ Determine Gain Margin and Phase Margin from the Bode Plot and comment whether the system is stable or not.	10	4	3

5.	Sketch the Polar Plot for the control system with transfer function given by, $G(s)H(s) = \frac{1}{s^2(1+s)(1+2s)}$ Comment on the System Stability from the Polar Plot by Computing Gain Margin and Phase Margin.	10	4	
*****All the best *****				